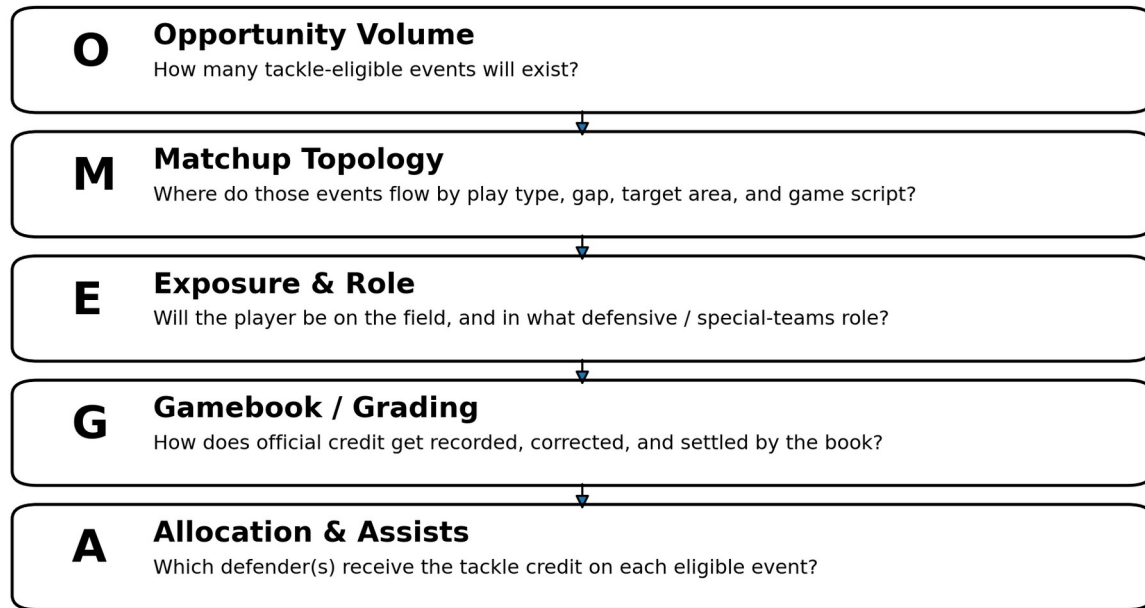


OMEGA Tackle Model

Research & Production Design Manual

A separate NFL defensive-player tackle engine for Tackles + Assists and related niche markets

OMEGA Tackle Model: five-layer causal architecture



Version 0.1 - Research specification

Prepared for the MODEL project - 2026-09-11

Core principle: model the football event and the official statistical credit independently; compare to the sportsbook only after the probability distribution exists.

Executive Thesis

OMEGA is not a rolling-average tackle model. It is a causal event model for how tackle credits are created, allocated, officially recorded, and finally settled by a sportsbook.

The defensive tackle market is attractive because the observable box-score statistic is produced by several distinct mechanisms that are usually collapsed into one number. A player can only receive a tackle credit if a tackle-eligible event occurs, if he is on the field in a role that can reach that event, if the event flows into his part of the defense, and if the official statistical crew assigns him credit. For some books, special-teams tackle credits are added as well. OMEGA separates those mechanisms rather than assuming that a player who averaged 8.1 combined tackles will continue to average 8.1.

The model is deliberately narrower than the broader NFL win-probability system. Its first production target should be full-game Tackles + Assists (T+A), beginning with high-snap linebackers and safeties. Solo tackles and assisted-tackle-only markets are later extensions because they are more sensitive to the scoring crew's solo/assist classification. The model can reuse team-state information from the NFL project - expected play volume, quarterback state, offensive style, personnel continuity - but it must maintain its own lineage, calibration, holdout discipline, market adapter, and Trust gates.

OMEGA is an acronym for the five causal layers that must all be right: Opportunity Volume; Matchup Topology; Exposure & Role; Gamebook / Grading mechanics; and Allocation & Assists. The thesis of this manual is that the possible edge is not one magic coefficient. It is the compound advantage from modeling details that generic prop systems can easily flatten: tackle-eligible plays instead of snaps, position-specific tackle funnels instead of opponent rank, teammate competition instead of isolated player averages, special-teams leakage, gamebook credit multiplicity, venue-year scoring tendencies, and book-specific settlement definitions.

What this manual does

- Defines what an NFL tackle and assist actually are under the league's statistical guidelines, including edge cases that matter for betting settlement.
- Translates NFL / GSIS taxonomy into a reproducible T+A target and highlights the naming traps between NFL box scores, nflfastR fields, and sportsbook labels.
- Designs a generative tackle model from play volume to tackle-eligible events to player credit allocation, rather than fitting a black-box outcome classifier first.
- Creates a research agenda for under-discussed effects: venue credit inflation, teammate cannibalization, tackle-concentration entropy, special-teams leakage, boundary-credit effects, role substitution offsets, and postgame correction risk.
- Defines a fail-closed market workflow in which PropsMadness captures prices while the independent probability engine remains isolated from the line.
- Provides an implementation roadmap, validation protocol, Trust framework, and regression tests for a production-quality OMEGA engine.

What this manual does not claim

It does not claim that any of the proposed "Omega Lab" hypotheses is already an exploitable edge. Some have historical support, some are informed by NFL scoring mechanics, and some are intentionally novel research hypotheses. They must survive chronological out-of-sample testing and then prospective 2026 tracking. The model should be willing to discover that a creative idea adds no predictive value and remove it.

Contents

1. Why tackles are a promising niche market
2. The official statistic: tackles, assists, combined, special teams and miscellaneous
3. Edge cases and exceptions that change credit
4. The sportsbook settlement contract
5. OMEGA architecture: Opportunity - Matchup - Exposure - Grading - Allocation
6. Data architecture and source hierarchy
7. O layer: opportunity volume and tackle-eligible events
8. M layer: matchup topology and the tackle funnel
9. E layer: exposure, role, rotation and special teams
10. G layer: gamebook mechanics, venue effects and stat corrections
11. A layer: teammate competition, credit multiplicity and allocation
12. Distribution engine, uncertainty and calibration
13. Omega Lab: niches within the niche
14. Market comparison, Trust and execution
15. Validation discipline and prospective proof
16. Build roadmap and extension design
- Appendix A. Feature dictionary
- Appendix B. Settlement and integrity regression tests
- Appendix C. Synthetic worked example
- Appendix D. Sources and research notes

1. Why Tackles Are a Promising Niche Market

NFL moneylines and spreads are highly liquid, heavily modeled, and rapidly corrected by sophisticated market participants. A defensive player tackle prop is a different forecasting problem. The book must price hundreds of player-level outcomes from a shared game state, often with lower limits and less two-sided liquidity. That does not guarantee inefficiency, but it creates more places where a projection can be wrong for a specific, identifiable reason.

Tackles are also unusually suitable for a structural model. They are counts, but they are not arbitrary counts. A tackle credit requires a chain of events: the offense runs a play; the play ends in a way that can create a defensive credit; a specific defender is on the field; the geometry and assignment of the play bring him into the event; and the official scorer records his involvement as a tackle or assist. That makes tackles closer to an opportunity-allocation process than to a pure “hot hand” statistic.

The NFL itself has moved in this direction. Next Gen Stats introduced Tackle Probability, a tracking-based model that estimates each defender’s chance of making a tackle every tenth of a second using multiple features. The public implementation details are limited, but the existence of that official metric validates the core idea that raw tackle counts omit the opportunity context surrounding the play. [8]

OMEGA cannot replicate the NFL’s proprietary live tracking feed. It does not need to. Public play-by-play, participation, snap-count and roster data can create a lower-resolution but highly interpretable opportunity model. The objective is not to out-track Next Gen Stats; it is to price a sportsbook settlement target more accurately than a simple player-average plus opponent-rank approach.

The first narrow target

Initial production scope: full-game Tackles + Assists for high-snap linebackers and safeties. No touchdowns, no sacks-only props, no first-tackle markets, no same-game parlay optimization, and no market-derived xT+A.

This scope is intentionally restrictive. Linebackers and safeties offer the clearest path because their tackle production is heavily connected to defensive snap share, run/pass mix, short-area completions, scrambles, and role. Rotational defensive linemen and edge players can be added later; their low counts, package-specific snaps, and sack-driven tackle credits create a different distribution. Cornerbacks can also become valuable, particularly against high target or screen volume, but boundary and coverage effects should be understood before they are promoted.

Why T+A before solo tackles

The NFL’s official defensive scoring framework allows a primary defender to receive a tackle while another receives an assist, or two equally responsible defenders to each receive an assist. For a T+A wager, a player receives one combined credit either way. That means some solo-versus-assist subjectivity cancels out. It does not remove scorer subjectivity entirely - a scorer can still decide whether a second defender deserves any credit - but T+A is less exposed to classification noise than a solo-tackle-only market. [1]

2. The Official Statistic: Tackles, Assists, Combined, Special Teams and Miscellaneous

A tackle prop cannot be modeled correctly until the target statistic is defined exactly. The NFL’s Guide for Statisticians says a tackle is credited when one defender is solely or primarily responsible for stopping the ball carrier. It explicitly states that a sack is also a tackle and that a forced fumble is recorded as either a tackle or an assist depending on the play. The same guide describes two assist situations: a second defender materially helps finish a stop that is primarily made by another defender, or two defenders contribute equally and each receives an assist. [1]

The NFL’s GSIS cumulative-statistics schema is equally important because it separates several concepts that casual box-score language can blur. The defensive player record contains Tackles, Assists and Combined, where Combined is defined as Tackles + Assists. It also maintains separate SpecialTeamsTackles, SpecialTeamsAssists, and MiscellaneousTackles fields. [2]

Term	Operational meaning for OMEGA	Why it matters
Defensive tackle	Official defensive tackle credit in the GSIS defensive record.	Primary input to Combined.

Term	Operational meaning for OMEGA	Why it matters
Defensive assist	Official assist credit in the GSIS defensive record.	Also contributes one unit to Combined.
Combined	GSIS Tackles + Assists.	Closest official concept to a normal T+A prop before book-specific additions.
Special-teams tackle / assist	Separate GSIS fields for tackle credits on special-teams plays.	Some books explicitly include them in T+A settlement.
Miscellaneous tackle	Tackle recorded when the player's team started the play with possession.	Do not silently mix into a defensive T+A prop unless the book says so.
Sack	A sack is also a tackle under the defensive scoring guidelines.	Pass-rush plays create tackle credits but funnel them to different positions.

The nflfastR naming trap

nflfastR exposes three separate defensive tackle-related player statistics: `def_tackles_solo`, `def_tackles_with_assist` ("tackles where another player assisted"), and `def_tackle_assists` ("assist to another player's tackle"). [4] This distinction is more granular than a typical sportsbook interface. A naïve implementation that adds only solo tackles plus tackle assists can undercount official Combined because it can omit tackles where the player was the primary tackler and another defender assisted.

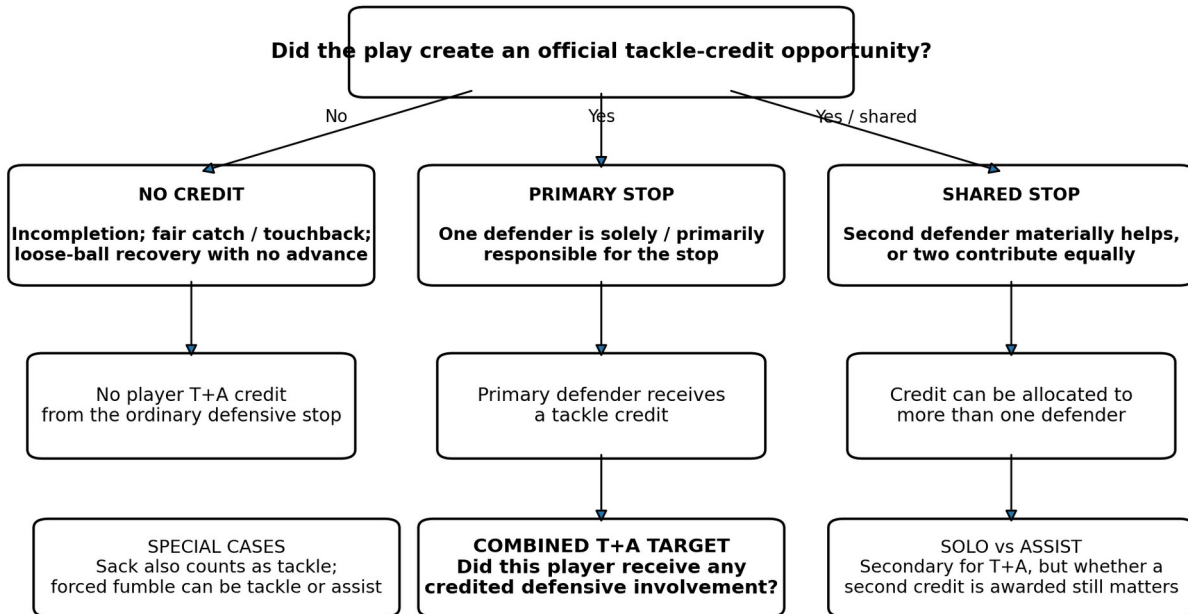
Candidate defensive Combined reconstruction = solo tackles + tackles-with-assist + tackle assists

That formula must be regression-tested against official GSIS Combined totals before it is allowed into production. It is an engineering hypothesis based on the documented field semantics, not a reason to trust an unverified aggregation blindly.

Credit is not contact

The NFL rules are written around responsibility for ending the play, not around a simple collision counter. A defender can receive a tackle when the ball carrier goes out of bounds to avoid or because of that defender. Conversely, physical proximity does not guarantee an assist if the second defender did not materially hasten the end of the play or prevent additional yardage. [1] This distinction is central to OMEGA: the target is the official credit process, not "who touched the runner."

Official-credit logic that OMEGA must reproduce



Implementation rule: model the official statistical event, not merely physical contact.

Figure 1. OMEGA models official statistical credit, not merely physical contact.

3. Edge Cases and Exceptions That Change Credit

The apparent simplicity of a tackle count hides a long list of state transitions that decide whether a defensive credit can exist. The model should encode these as an event ontology and verify the ontology against official play-by-play rather than merely count defensive snaps.

Situation	Expected credit treatment	OMEGA implication
Ordinary rush stopped in field	One or more defensive credits may be recorded.	Core tackle-eligible event.
Completed pass stopped in field	One or more defensive credits may be recorded.	Completion depth and YAC shape drive positional funnel.
Ball carrier goes out because defender impedes progress	Defender may receive tackle even without conventional takedown.	Boundary plays need an out-of-bounds credit model.
Incomplete pass	No ball carrier stop.	Dropback exists, tackle opportunity usually does not.
Sack	Sack is also a tackle.	Creates a tackle event concentrated among pass rushers.
Forced fumble during stop	Forcer receives tackle or assist depending on play.	Fumble events cannot simply be removed from tackle modeling.
Loose-ball recovery with no clear attempt to advance	NFL guide says no tackle should be credited.	Exclude as tackle opportunity.
Interception and return	The defending team becomes offense; return tackles are separately categorized.	Do not confuse miscellaneous return tackles with defensive T+A.
Kick / punt return stopped	Special-teams tackle / assist.	Can count in sportsbook T+A depending on settlement contract.
Fair catch / touchback	No returner stop.	No special-teams tackle opportunity.
QB spike	Incomplete pass event.	No tackle opportunity.
QB kneel / no-play / nullified play	Stat treatment must follow official PBP outcome.	Empirically audit; do not count “snaps” that generate no tackle statistic.
Overtime	If book rules include OT, all additional eligible events matter.	Simulate an overtime tail.

Replay and postgame corrections

The target can change after the stadium crew's first entry. NFL Fantasy publishes official statistical corrections released by the NFL League Office and Elias Sports Bureau. Examples from 2025 include changes that moved one unit from tackle to assist without changing a player's combined total, and other corrections that added or removed assists outright, changing Combined. [3]

This yields two correction classes that OMEGA should track separately: composition corrections, where solo/assist mix changes but T+A is unchanged, and combined-changing corrections, where the wager's settlement count itself changes. The first matters heavily for solo and assists-only markets; the second matters even for T+A.

Penalty / no-play handling

nflfastR explicitly identifies no_play, qb_kneel, qb_spike and other play types, and exposes penalty-nullified play context. [5] OMEGA should build its historical labels from finalized official defensive credits and then learn which raw play states actually produce those credits. This is safer than hand-authoring a brittle list of every penalty exception. A “raw defensive snap” is not automatically a tackle opportunity.

4. The Sportsbook Settlement Contract

A model can be statistically brilliant and still lose if it predicts the wrong settlement definition.

The first OMEGA production artifact should therefore be a settlement contract, not a neural network. FanDuel's published American Football house rules in at least one U.S. jurisdiction explicitly state that Tackles + Assists includes those recorded on both Defense and Special Teams. The same rules say overtime counts unless stated otherwise. [11] That means a defense-only model can be systematically mis-specified even when its defensive projection is perfect.

Illustrative FanDuel target = Defensive Tackles + Defensive Assists + Special-Teams Tackles + Special-Teams Assists

That equation must be validated against actual settled markets and the current house rules in the user's jurisdiction before production. House rules can vary by state and can change. OMEGA should archive the rule text and a hash alongside each market adapter version.

Why this creates a genuine niche-within-a-niche

A player's special-teams contribution is often small relative to an every-down linebacker's defensive total, but half-point prop markets can be decided by a single credit. The effect can be proportionally much larger for safeties, reserve linebackers and role-changing players. It also creates an offset that simple projection systems can miss: when a backup becomes a full-time defensive starter, his defensive snap expectation can rise while his special-teams workload falls. The correct total T+A change is therefore not "old tackles plus extra defensive snaps." The two opportunity streams must be modeled jointly.

Book-specific taxonomy

DraftKings and other platforms also list separate Solo Tackles, Assisted Tackles, and Tackles + Assists categories, but availability and settlement details must be verified from each operator's current rules rather than inferred from labels. [12] OMEGA should treat the sportsbook market name as an opaque identifier until a settlement mapping has been validated.

Contract field	Required value before market can be VERIFIED
Book / jurisdiction	Exact operator and applicable ruleset.
Market name	Exact UI/API label, e.g. Tackles + Assists.
Defense included?	Yes/No, proven from rules or settled examples.
Special teams included?	Yes/No, proven.
Miscellaneous tackles included?	Yes/No, proven.
Overtime included?	Yes/No, proven.
Participation void rule	Exact snap/appearance condition.
Correction cutoff	Whether later official corrections can alter settlement.
Official stat provider	League / provider used for final grading.
Contract hash	Immutable hash of the captured rule text / adapter.

5. OMEGA Architecture: Opportunity - Matchup - Exposure - Grading - Allocation

OMEGA Tackle Model: five-layer causal architecture

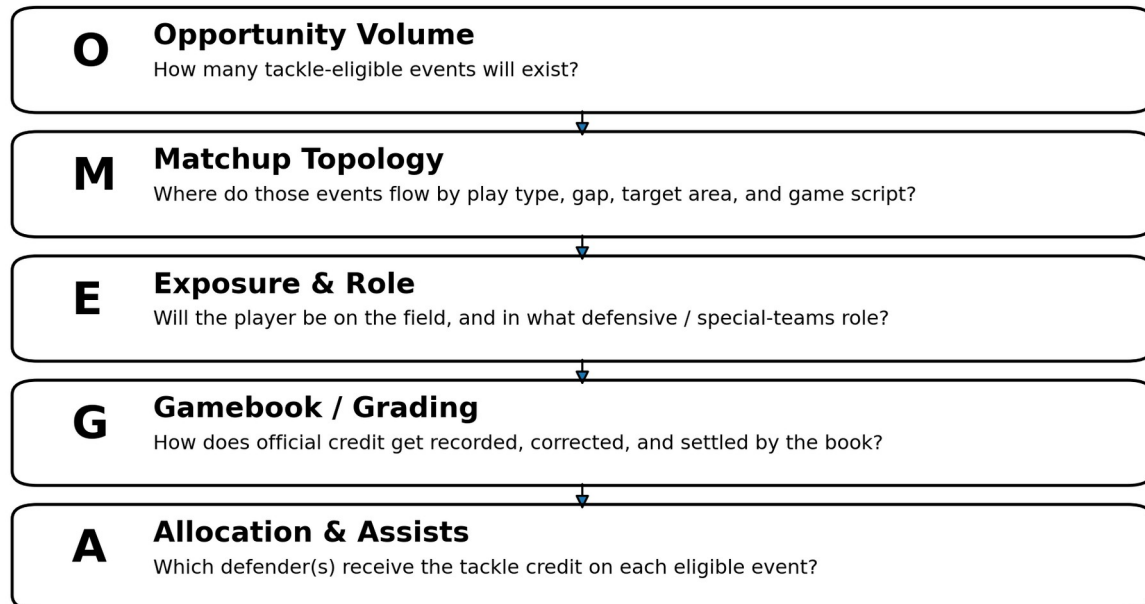


Figure 2. The OMEGA five-layer causal decomposition.

The model should be designed so that each layer can fail independently and can be audited. This makes it easier to discover where an apparent edge comes from and easier to block a wager when a critical input is unresolved.

O - Opportunity Volume

Forecast the number and type of tackle-eligible events created by the opponent. This is not the same as defensive snaps. Incompletions, touchdowns, no-plays, fair catches and touchbacks can consume play volume without creating a conventional tackle credit. OMEGA should forecast events such as rushes that end in a stop, completed passes that end in a stop, scrambles, sacks and returned kicks.

M - Matchup Topology

Forecast where those events flow: inside run versus edge run, short middle completion versus boundary completion, screen versus deep shot, scramble versus sack, run gap, formation and coverage. nflfastR provides run location, run gap, pass location, air yards, yards after catch, scramble and related fields; nflverse participation adds defenders in the box, personnel, route, pressure and the identities of players on the field. [5][7]

E - Exposure & Role

Forecast whether the player is on the field for each event family. Aggregate defensive snap share is only the first approximation. A linebacker can be a 95% player overall but leave the field in a particular dime package; a safety can move into the box after an injury; a reserve can gain defensive snaps and lose special-teams snaps. nflverse snap-count data exposes offense, defense and special-teams snaps and percentages. [6]

G - Gamebook / Grading

Model the official credit process and settlement mapping: whether the event generated one or multiple tackle credits, whether a second defender received an assist, whether a boundary exit was credited as a tackle, whether the venue-year exhibits a persistent residual tendency, and whether a postgame correction changed the official total.

A - Allocation & Assists

Allocate the event's tackle credit among the defenders who were actually on the field. A player's tackle average is partly a share of a team-level opportunity pool. Teammates compete for primary credit, but group tackles can create more than one combined credit. OMEGA therefore needs both competition and co-credit behavior.

6. Data Architecture and Source Hierarchy

The data design should mirror the project's existing trust philosophy: raw source capture is immutable; provider normalization is separate from modeling; market data is physically quarantined from probability generation; and every production probability carries source lineage.

Source	What OMEGA uses	Role / caveat
NFL GSIS Guide for Statisticians	Official tackle/assist rules and exceptions.	Authority for statistical semantics. [1]
NFL GSIS cumulative statistics	Tackles, Assists, Combined, SpecialTeamsTackles/Assists, Miscellaneous.	Settlement reconciliation. [2]
nflverse / nflfastR play-by-play	Play types, tackler IDs, assists, run/pass location, YAC, sacks, scrambles, no-play.	Historical event ledger. [4][5]
nflverse participation	Players on play, personnel, box count, routes, pressure, coverage.	Play-level exposure and matchup topology from 2016 onward. [7]
nflverse snap counts	Defense and special-teams snaps / percentages.	Role priors and postgame verification. [6]
NFL / team live roster and inactives	Pregame player availability.	Required live role gate; historical timestamps must be verified.
PropsMadness NFL	Book-level tackle lines/prices and potentially matchup metadata.	Market only; opaque proprietary composites are research/diagnostic until validated.
OddsPapi	No tackle props under current project plan.	Do not probe player props; game markets only.
NFL official stat corrections	Tackle/assist revisions after initial scoring.	Correction-risk and label finalization. [3]
Big Data Bowl / NGS research data	Tracking-based tackle opportunity research on limited historical samples.	Research challenger / geometry validation, not live production feed. [8][9][10]

Canonical identifiers

Every event should key to nflverse game_id + play_id + gsis_id. Player names are display fields only. The existing NFL project already treats gsis_id as the canonical player identity. OMEGA should continue that rule. Market player names

from PropsMadness must join through a normalized identity table, with fail-closed behavior on ambiguous names, trades, suffixes, or duplicated surnames.

Immutable ledgers

- OMEGA_EVENT_LEDGER: one row per finalized play, containing play type, tackle eligibility, official credited defenders, and source hashes.
- OMEGA_PLAYER_GAME: one row per defender-game with defense snaps, ST snaps, official T/A/ST counts, role state and final target.
- OMEGA_PREGAME_FEATURES: only information available before kickoff; no current-game snaps, tackles, or outcomes.
- OMEGA_MARKET_CAPTURE: separate market lineage with book, line, price, timestamp, settlement-contract ID and capture source.
- OMEGA_PROSPECTIVE_LEDGER: frozen model probability before market outcome and before postgame labels.

7. O Layer: Opportunity Volume and Tackle-Eligible Events

The biggest conceptual error in tackle handicapping is treating every defensive snap as equivalent. OMEGA should forecast the count of “tackle-eligible events” (TEOs): plays whose final statistical outcome can reasonably generate a defensive or special-teams tackle credit under the settlement contract.

$$\text{Expected T+A}_i = \text{sum over event classes [Expected events}_{\text{class}} \times P(\text{player credited} \mid \text{event}_{\text{class}})] + \text{expected special-teams credits}$$

Event classes

Class	Key drivers	Primary tackle funnel
Inside rush	rush rate, run gap, box count, OL/RB style, game script	DL / off-ball LB
Edge rush	run location, perimeter blocking, formation	LB / S / CB
Short middle completion	completion rate, route mix, YAC, zone usage	LB / S
Short boundary completion	target location, route, YAC, sideline	CB / S / LB
Deep completion	deep rate, completion probability, touchdown rate	S / CB; fewer events but large field geometry
Screen / swing	route tag, pressure response, YAC	LB / S / CB
QB scramble	QB scramble tendency, pressure, man/zone	LB / S
Sack	dropbacks, pressure/sack propensity	EDGE / DL / blitz
Kick return	kick return rate, touchback rate, coverage role	ST coverage players
Punt return	punt/fair-catch/return mix, coverage role	ST coverage players

Drive survival multiplier

Pace alone is not enough. A fast offense that goes three-and-out repeatedly can create fewer sustained tackle opportunities than a slower offense that strings together first downs. OMEGA should model expected drives and expected plays per drive jointly. Completion rate, third-down conversion, sack rate, turnover rate, explosive touchdowns, and penalty/no-play frequency all affect the number of actual credit-producing events.

The incompleteness and touchdown taxes

A pass-heavy opponent does not automatically create tackle overs. Incompleteness are dropbacks with no receiver tackle. Long touchdowns end without a defensive stop at the end of the scoring play. A high-volume short-passing offense can be far more tackle-friendly to linebackers and safeties than an equally pass-heavy vertical offense. This is why “opponent pass attempts rank” is insufficient; completion shape matters.

Overtime tail

If the settlement contract includes overtime, the count distribution should include the probability of extra play volume. Overtime is rare, but on a half-point line a few extra tackle-eligible events can matter. OMEGA should simulate overtime as part of the game distribution rather than add a fixed average to every projection.

8. M Layer: Matchup Topology and the Tackle Funnel

A tackle funnel is the conditional distribution of defensive credit by position and player role given the kind of offensive play that occurred. OMEGA should learn these funnels empirically rather than assign them from intuition.

For example, an inside zone run stopped after three yards may funnel credit toward a defensive tackle and inside linebacker. A shallow cross can funnel credit to a linebacker or safety. A deep boundary completion can funnel credit to a corner or safety. A sack concentrates credit in the pass rush. The same number of offensive plays can therefore produce very different tackle distributions across defenders.

Opponent Tackle Funnel Fingerprint

For each offense, construct a rolling, shrunk fingerprint of how its successful and unsuccessful plays distribute tackle credits across defensive positions and zones. The fingerprint should be conditioned on play class rather than being a raw “tackles allowed to linebackers” average. A team can appear linebacker-friendly merely because it ran many plays; OMEGA should separate volume from allocation.

Candidate fingerprint components

- Rush location: left / middle / right and gap: end / guard / tackle.
- Pass location: left / middle / right; short / deep; air yards and YAC.
- Route family where available: screen, shallow cross, slant, swing, angle, out, post, go and related groups.
- Formation and personnel: shotgun, singleback, receiver / tight-end / back counts.
- Defenders in box and defensive personnel package.
- Pressure / time-to-throw / scramble state.
- Down, distance, field position, red zone, two-minute situation and score state.

A subtle pass-rush interaction

Pressure can lower linebacker/safety tackle volume even when opponent dropbacks rise. Pressure creates sacks, which are tackle credits concentrated among pass rushers, and incompletions, which create no receiver tackle at all. The same matchup can therefore be “high-volume passing” but a poor T+A environment for second-level defenders. OMEGA should let dropback outcomes determine the tackle funnel rather than treat dropback count as a direct tackle proxy.

Boundary-credit effect

Because the NFL allows tackle credit when a defender causes a ball carrier to step out of bounds, boundary usage can create a distinct official-credit channel. [1] OMEGA should test whether specific positions, players, offenses or venue-years show stable residual tackle-credit rates on out-of-bounds endings after controlling for play type and location. This is a hypothesis, not a production feature until it improves out-of-sample prediction.

9. E Layer: Exposure, Role, Rotation and Special Teams

Exposure is the bridge between team-level opportunity and player-level probability. The best matchup in the league is irrelevant if the player only plays 52% of defensive snaps. Conversely, a role promotion can create a market lag before rolling tackle averages catch up.

Role state instead of one snap-share number

OMEGA should represent role as a state vector: expected overall defensive snap share, expected early-down share, passing-down share, box / slot / deep role where observable, blitz involvement, goal-line package participation, and special-teams exposure. nflverse participation provides players on each play plus personnel, route and coverage information from 2016 onward, enabling historical role inference far beyond aggregate snap percentage. [7]

Role substitution offset - an under-discussed trap

When a backup LB becomes a starter, defensive snaps may rise while special-teams snaps fall. For a book whose T+A includes special teams, the net projection change can be smaller than a defense-only model expects.

This should be modeled explicitly with a joint role transition. The same player cannot simply add 40 defensive snaps while retaining the old 70% special-teams workload if the coaching staff normally removes starters from coverage units. Historical transitions can be grouped by position, team, and type of promotion to estimate the expected offset.

Rotation entropy

Some defenses use stable every-down linebackers; others rotate packages aggressively. Define a team-package rotation entropy from play-level defender participation. High entropy means role uncertainty is intrinsically larger, even when the nominal depth chart is clear. This can become an uncertainty input rather than a direct mean adjustment: identical xT+A point estimates should receive different confidence if one role is stable and the other is package-dependent.

Pregame gating

- Player identity must resolve to `gsis_id`.
- Active/inactive status must be verified close to kickoff.
- Recent defensive and ST snap histories must be fresh.
- Any major teammate absence that changes role must be represented in the role state or trigger a WATCH/BLOCKED status.
- If the player's role is genuinely new and historical transition analogs are weak, increase uncertainty rather than pretending to know the exact snap share.

10. G Layer: Gamebook Mechanics, Venue Effects and Stat Corrections

T+A is less scorer-sensitive than solo tackles, but it is not scorer-free. The official rules require judgment about who was primarily responsible, whether a second defender materially contributed, and whether an out-of-bounds ending was caused by a defender. The NFL's own guide concludes that situations not fully covered should be resolved with common sense, which necessarily leaves a human classification layer. [1]

Historical venue / stat-crew effect: test, do not assume

Historical IDP research has reported large venue differences in the way crews awarded solos and assists. [13] Those studies are old enough that OMEGA should not import their multipliers. The correct approach is to re-estimate a venue-year effect on modern data and shrink it aggressively toward zero. The model should only retain the effect if it persists out of sample after controlling for offensive play mix, defensive personnel and player roles.

The key identification trick

Do not estimate "stadium generosity" from raw home-team tackles. That confounds the home defense's scheme and personnel with the venue. Instead, use both teams and especially visiting defenders as a quasi-control. A hierarchical model can estimate a venue-year residual for the number of official combined credits generated per tackle-eligible event, while team, opponent, position, play class and season absorb football differences.

Credit Inflation Factor_{v,y} = observed combined credits / expected combined credits after football controls

A second model can estimate the solo-versus-assist mix. That second quantity is more relevant to solo and assists-only markets; the first is directly relevant to T+A.

Stat-correction susceptibility

NFL official correction pages provide a unique data set for measuring label instability. [3] OMEGA should create a Correction Delta Ledger: original total, corrected total, correction type, position, venue, week, and time to correction. The objective is not necessarily to predict a correction for betting; it is to know how much settlement noise sits around a half-point line and whether some venues or stat categories are more correction-prone.

Examples of why the distinction matters

A correction that changes a linebacker from 6 tackles / 4 assists to 5 tackles / 5 assists leaves Combined at 10 and should not affect a T+A wager, even though it matters to solo and assist markets. By contrast, an official correction that raises assisted tackles from 4 to 6 changes Combined by two and can flip T+A settlement. Both types appear in NFL correction records. [3]

11. A Layer: Teammate Competition, Credit Multiplicity and Allocation

The player-level problem is not “Will this play be a tackle?” It is “Which defender or defenders receive the official credit?” That makes tackle production a constrained allocation problem.

A rule-aligned allocation model

For each tackle-eligible event j , OMEGA can model the number of credited defenders K_j and then allocate those credits among defenders on the field. The NFL guide describes ordinary single-primary and two-defender shared outcomes. nflfastR’s play-by-play schema exposes multiple tackle and assist player slots, so the implementation should not hard-code a two-credit maximum until the historical raw data are audited. [1][5]

$P(K_j \mid \text{play class, traffic, venue-year, game state}) \rightarrow \text{allocate } K_j$
credits among eligible defenders

This structure is preferable to eleven independent Bernoulli models because it respects the fact that defenders compete for the same event. It also naturally handles co-credit: a linebacker and safety can both receive one combined credit on the same play.

Tackle concentration entropy

Define each defender’s share s_i of team combined defensive credits over a stable window and compute $H = -\sum(s_i \log s_i)$. A low-entropy defense funnels a large fraction of tackle credits through a small number of players; a high-entropy defense distributes them more broadly. Entropy can be estimated overall and by play class. The hypothesis is that low-entropy roles produce more stable player-level tackle shares and larger redistribution after a central player is inactive.

Teammate cannibalization network

Create a co-credit network where an edge between defenders i and k is weighted by the frequency with which both receive a credit on the same play. Add a competition edge based on how often one receives the primary credit when both are on the field. This network can answer questions that player averages cannot: when LB1 is inactive, does LB2 inherit the missing share, does a safety inherit it, or does the defense change package and distribute the opportunities?

Credit elasticity

Estimate how a player’s tackle-credit share changes as team tackle-eligible volume rises. Some every-down linebackers may absorb additional volume nearly proportionally; others may have a stable role whose share shrinks when the opponent moves into specific pass-heavy packages. A player with high positive credit elasticity can benefit more from a high-play environment than a simple average-share projection implies. This is a research hypothesis to test, not a preloaded coefficient.

12. Distribution Engine, Uncertainty and Calibration

OMEGA as a generative game simulator

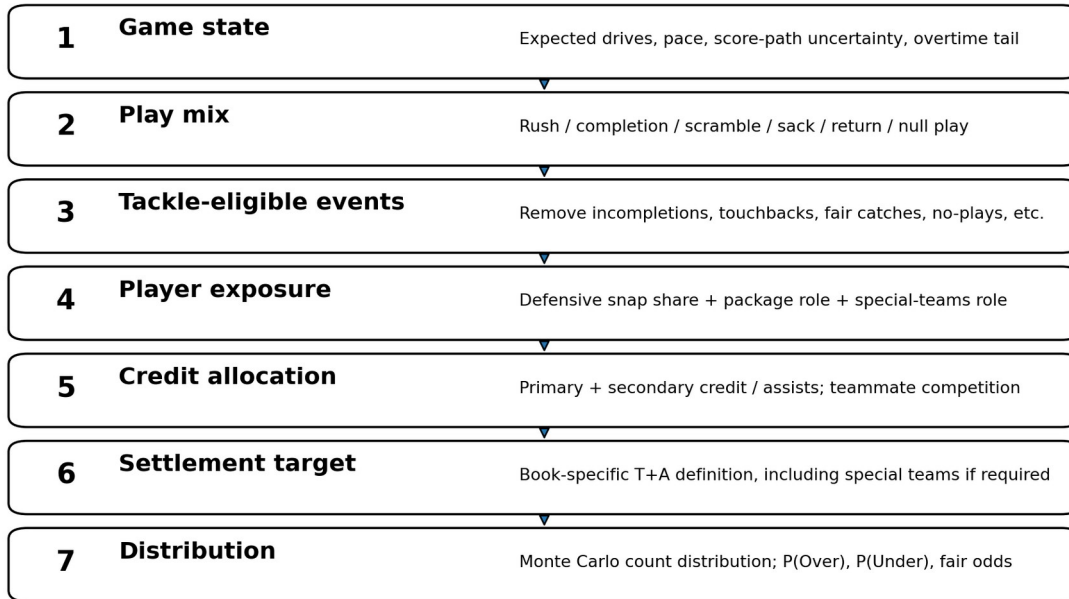


Figure 3. The preferred OMEGA generative workflow.

A point estimate is not enough to price a prop. The engine must output a full integer count distribution because the same xT+A can imply different Over probabilities under different variance and role uncertainty.

Why plain Poisson is not enough

A Poisson model assumes variance equals the mean and independent event arrivals. Tackle counts violate both assumptions. Team play volume varies by game script; player role can change during a game; events are clustered within drives; shared tackle credit induces dependency; and an early injury creates a low-count tail. The initial baseline can use an overdispersed negative-binomial or Poisson-mixture model, but the target architecture should be Monte Carlo game simulation.

Monte Carlo state draws

1. Draw a game-level play/drive environment from the pregame team-state distribution.
2. Draw event-class counts conditional on game state: inside runs, boundary runs, short-middle completions, screens, sacks, scrambles, returns, and so on.
3. Draw player defensive and special-teams exposure from the role-state distribution.
4. For each tackle-eligible event, draw the number of official credits and allocate them among defenders on the field.
5. Add settlement-specific special-teams credits, overtime if applicable, and any approved venue-year grading effect.
6. Aggregate the player count and repeat enough times to estimate $P(T+A > \text{line})$, $P(T+A < \text{line})$, median, tails and fair American odds.

Uncertainty budget

Uncertainty source	Model response
Role / snap share	Widen role-state mixture; block if active/inactive unresolved.
Opponent play volume	Game-state / drive distribution.
Play-type mix	Hierarchical team tendencies with season-stage shrinkage.
Player tackle share	Player + role + position shrinkage.
Teammate availability	Scenario mixture until inactives verified.
Venue scorer effect	Strong shrinkage; zero if not demonstrably stable.
Special-teams role	Separate exposure distribution, correlated with defensive role.

Uncertainty source	Model response
In-game injury	Optional generic availability tail; never fake player-specific injury hazard without evidence.

Calibration targets

For count distributions, evaluate more than hit rate. OMEGA should track log score / negative log likelihood for the full count distribution, calibration of Over probabilities by line, Brier score for over/under events, mean absolute error of count, coverage of prediction intervals, and probability integral transform diagnostics. Market performance comes later and must never be the criterion used to choose football features.

13. Omega Lab: Niches Within the Niche

The ideas below are research hypotheses. Their value is that each has a falsifiable mechanism, a data path, and a reason it could be underrepresented in simple tackle projections.

Hypothesis	Mechanism	Test / falsification
1. Special-teams leakage	Book T+A may include ST tackles; reserve roles can contribute meaningful tail.	Reconcile settled book totals to GSIS defense + ST; compare defense-only vs combined model OOS.
2. Role substitution offset	New starter gains defensive snaps but often loses ST snaps.	Historical transitions: delta defense snaps vs delta ST snaps and delta T+A.
3. Tackle concentration entropy	Some schemes funnel credits to a small set of defenders.	Measure stability of player tackle share and redistribution after absences.
4. Teammate cannibalization network	Credit is allocated among defenders, not independently generated per player.	Pairwise on-field credit shares; injury/absence natural experiments.
5. Venue credit inflation	Human scoring may alter probability a second defender receives credit.	Hierarchical venue-year residual after controlling football factors; visiting defenders as quasi-control.
6. Boundary-credit effect	Out-of-bounds endings can still earn tackle credit.	Model credit rate on OOB plays by position/location/venue, OOS.
7. Completion-shape funnel	Short middle/YAC completions generate different LB/S opportunities than deep boundary throws.	Event-class position credit matrix and opponent fingerprint.
8. Pressure / sack tax	Pressure creates incompletions and sacks, reducing LB/S reception tackles while shifting credits to rushers.	Condition tackle share on pressure and sack outcomes.
9. Drive survival multiplier	First downs and completions increase future tackle opportunities; pace alone misses this.	Forecast TEOs with drive model vs raw plays/game.
10. Garbage-time rotation	High play volume can coincide with starter rotation in lopsided games.	Score-state $\times$ snap-share response by player role.
11. Overtime option value	Full-game props include OT at some books.	Simulated close-game OT tail vs no-OT distribution.
12. Correction susceptibility	Some types of credits are more likely to be revised postgame.	NFL correction ledger by venue/position/stat and combined-changing rate.
13. Market latency after inactives	Niche player lines may update slower than role projections after late news.	Prospective timestamped line movement around inactive announcements.
14. Key-line convexity	Small $xT+A$ changes near discrete count mass can cause large probability moves.	Compare delta xTA to delta $P(\text{over})$ across integer lines and dispersions.
15. Assist coalescence	Traffic-heavy events may create two official credits more often than open-field stops.	Estimate K_j credit multiplicity by play class, box count and position density.

A deeper angle: separate “getting there” from “getting credit”

The Big Data Bowl and NFL Tackle Probability work emphasize physical opportunity and tackling efficiency. [8][9][10] OMEGA’s betting-specific extension is to add a second layer: even if a defender is physically in the tackle opportunity, what is the probability the official gamebook gives him a credit? That distinction is especially important for group stops, boundary endings, and second-arriving defenders. We can approximate the physical opportunity layer from participation and play topology, then learn an official-credit conversion layer from GSIS labels.

A deeper angle: credit topology as a team fingerprint

Instead of treating player tackle rates as independent, represent each defense as a dynamic credit graph. Nodes are defenders; node weights are role-adjusted tackle shares; edges represent co-credit and competition. Coaching changes, injuries and package changes modify the graph. A prop edge can then arise from a graph transition before the player’s rolling average reflects it. This is potentially more informative than “Player X averaged 7.2 tackles in the last five.”

A deeper angle: model the book's exact count, not a fantasy statistic

The separation of defensive, special-teams and miscellaneous tackles in GSIS, combined with book-specific settlement language, suggests a practical source of model error in the market ecosystem: projections may be built from generic fantasy or box-score fields that do not perfectly match the sportsbook's settlement formula. OMEGA should never assume this happens at a book, but it should prove its own settlement reconstruction first and then test whether observed market errors cluster where taxonomy matters most.

14. Market Comparison, Trust and Execution

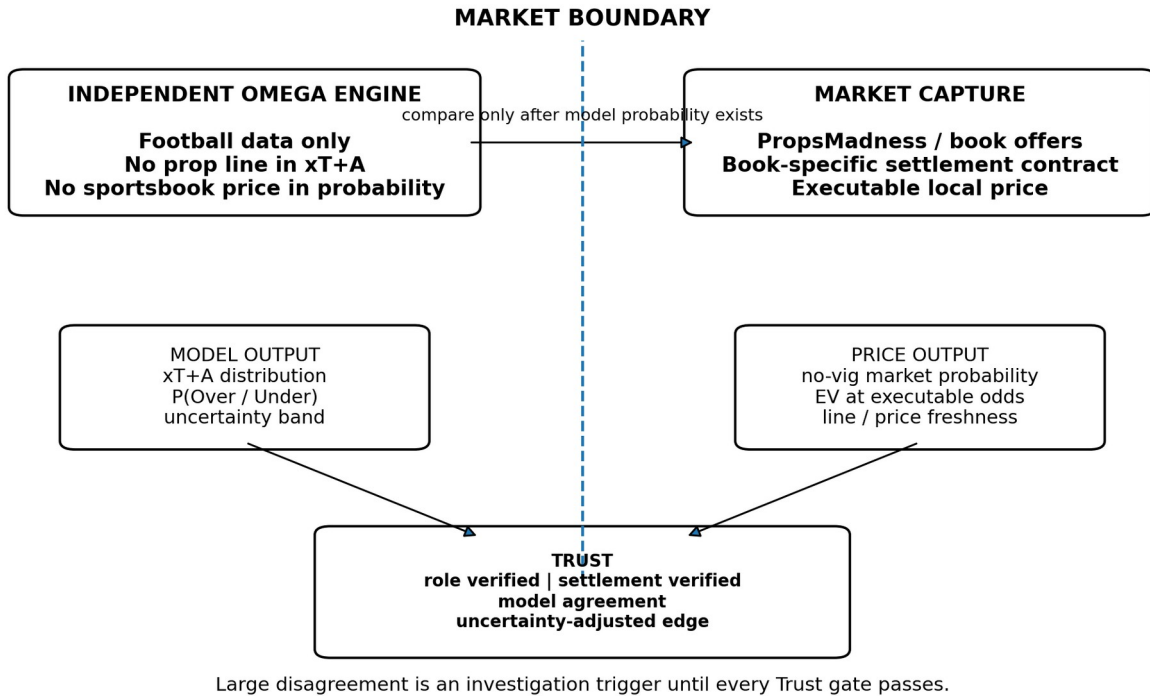


Figure 4. Market data remains downstream of the independent OMEGA probability.

The current project's OddsPapi plan does not include player props, so OMEGA must not spend OddsPapi requests trying to fetch tackles. PropsMadness is the natural price-capture candidate because the existing extension already interacts with it and its NFL UI exposes book-level player markets. Provider discovery should inspect the underlying network payload and preserve raw JSON; it should not screen-scrape pixels if a structured endpoint is available.

Market math

$$p_model = P_OMEGA(T+A > line)$$

$$p_market = \text{no-vig probability from the book pair when both sides are available}$$

$$edge_pp = p_model - p_market$$

$$EV = p_model \times \text{profit_if_win} - (1 - p_model)$$

For one-sided markets, OMEGA can compute executable EV from the offered price but should not pretend it has a no-vig market probability. Such rows are WATCH until a reliable two-way reference exists.

Trust gates

Gate	VERIFIED requirement
Identity	Canonical player / game / team resolved.
Settlement	Book contract validated for this market and jurisdiction.
Availability	Player active; key teammate inactives resolved.
Role	Defensive and ST role within model-supported uncertainty.
Source freshness	Market and availability timestamps within configured limits.

Gate	VERIFIED requirement
Model calibration	Player/position cohort belongs to calibrated scope.
Model agreement	Primary and simpler benchmark point in same direction or disagreement is investigated.
Uncertainty-adjusted edge	Conservative probability bound still clears required threshold.
Anomaly	Extreme model-market disagreement triggers research review, not automatic promotion.

Price shopping matters more in niche props

Player props often carry wider hold than main game markets. A modest probability edge can disappear if the executable price is poor. OMEGA should therefore track line and price separately. A player can be an Over at 6.5 -110 and a pass at 7.5 -120. The output must always be tied to an exact line, exact price and exact timestamp.

Prospective CLV ledger

Every candidate should record model probability, line, offered price, sharp/consensus reference if available, and later closing line/price. The prospective ledger should distinguish model correctness from execution quality: did the line move toward OMEGA, did the player hit, and was the price still positive EV after movement? Market results are evaluation, not training feedback to the frozen independent engine during the prospective phase.

15. Validation Discipline and Prospective Proof

The tackle engine must earn authority independently of the moneyline model. The fact that the NFL team model produced a directional holdout pass does not validate a defensive-player prop model. OMEGA needs its own chronological design.

Recommended historical protocol

1. Build and audit official tackle labels and settlement reconstruction before fitting player probabilities.
2. Use older seasons for feature discovery and role priors; never random-shuffle games across time.
3. Use chronological folds so role, scheme and scoring trends are always learned from the past.
4. Freeze event ontology, feature families, shrinkage process, distribution family, settlement target and Trust rules before final historical confirmation.
5. Treat 2025 as historical confirmation only, not the sole proof of edge; the project has already inspected some 2025 correction examples. The decisive evidence should be prospective 2026 performance and closing-line behavior.
6. Do not use sportsbook prices to select football features. Market data joins only after the independent candidate is frozen.

Metrics hierarchy

Layer	Primary metrics
Count model	Log likelihood / log score; MAE; interval coverage; calibration of count CDF.
Over/Under probability	Brier; log loss; reliability by probability bin and line bucket.
Role model	Snap-share MAE; package participation calibration; inactive/change-state coverage.
Opportunity model	TEO count error by event class and offense.
Allocation model	Player credit share log loss; top-credit accuracy; co-credit calibration.
Market evaluation	CLV, no-vig edge calibration, ROI with confidence intervals, price sensitivity.

Ablation discipline

Every creative Omega Lab feature must survive an ablation. Examples: remove venue effect; remove special teams; replace tackle-funnel topology with opponent position rank; remove teammate network; remove route/YAC features; remove game-state simulation. If a feature does not improve proper scoring rules out of sample or does not improve calibration, it leaves the production model even if the story sounds compelling.

Prospective success definition

A “market edge” is not declared because the model is interesting or because a few bets win. OMEGA earns that label only after pre-registered prospective probabilities are calibrated, show positive closing-line value or equivalent price improvement, and produce positive EV/ROI with enough volume to distinguish signal from noise.

16. Build Roadmap and Extension Design

OMEGA should be a separate engine under the NFL project, not a patch inside the frozen moneyline model. A suggested lineage is `nfl-tackle-omega-v0.1-event-credit-research-2026-09-11`, with calibration state `UNFIT\_RESEARCH` until historical tests are complete.

Phase	Deliverable	Promotion gate
T0 - Contract Audit	Book settlement ontology; GSIS reconciliation; PM tackle endpoint discovery.	100% understood target on sampled settled games; no ambiguous market mapping.
T1 - Event Ledger	2016+ tackle-credit event table with play / player / role joins.	Coverage and identity audit; finalized stats reconcile to GSIS/player totals.
T2 - Opportunity Baseline	Expected TEOs + simple role-adjusted xT+A baseline.	Chronological calibration beats player rolling-average baseline.
T3 - Tackle Funnel	Play-class / position allocation model using participation and PBP topology.	OOS improvement in count log score and O/U Brier.
T4 - Allocation Network	Teammate competition / co-credit model; role transition logic.	Ablation proves incremental value.
T5 - Grading + ST	Venue-year grading challenger; stat-correction ledger; special-teams sidecar.	Only components with OOS support survive.
T6 - Frozen Historical Candidate	Full distribution, calibrated and frozen.	No market input used; historical confirmation passes.
T7 - Prospective Alpha	Live 2026 board, PM prices, Trust, CLV ledger.	Sufficient prospective evidence before VERIFIED status / meaningful sizing.

Suggested extension surface

- `NFL | TACKLES` mode separate from NFL moneyline.
- `SYNC TACKLE DATA` - refresh live roster/inactive/role context; no market calls.
- `BUILD OMEGA` - generate independent xT+A distributions from football data only.
- `REFRESH TACKLE EDGES` - capture PropsMadness lines/prices and compare downstream; OddsPapi player-prop requests remain zero.
- `TACKLE TRUST AUDIT` - print identity, role, settlement, freshness, probability, no-vig edge, EV, uncertainty and reason codes.
- `COPY OMEGA AUDIT` - immutable text artifact for postgame / CLV review.

Production output card

Each player-line pair should expose enough decomposition that a human can understand the signal. A proposed card contains: player and role; exact market line and price; xT+A mean/median; P(Over) and P(Under); expected defensive snaps; expected ST snaps; tackle-eligible event count; player credit share; ST contribution; venue/grading adjustment if active; uncertainty interval; benchmark probability; no-vig market probability; executable EV; market freshness; role verification; and Trust status.

Appendix A. Feature Dictionary

Family	Candidate features / states
Game volume	expected drives, plays/drive, pace, no-huddle, third-down conversion, turnover rate, score-state distribution, OT probability
Tackle eligibility	rush stops, completions ending in field, scrambles, sacks, fumble advances, kick returns, punt returns, out-of-bounds credit class
Rush topology	run location, run gap, shotgun/under-center, personnel, box count, down/distance, field position
Pass topology	pass location, air yards, YAC, route, pressure, time-to-throw, screen/shallow/deep class, completion probability
Player role	def snap share, package share, box/slot/deep proxy, blitz/pressure involvement, ST snap share, rotation entropy
Player skill / credit	historical credited involvement per eligible event, position prior, player random effect, open-field vs traffic split
Team allocation	tackle concentration entropy, teammate co-credit network, competition share, absence redistribution
Grading	venue-year credit inflation, solo/assist mix, OOB credit rate, correction susceptibility
Live availability	active/inactive, role change, key teammate absence, current depth state
Market downstream	book, market label, line, over/under price, timestamp, settlement contract, no-vig p, EV; never model inputs

Shrinkage principles

Player tackle efficiency should shrink toward position/role priors when the sample is small. Opponent tackle funnels should shrink toward league means early in the season. Venue-year grading effects should shrink especially hard because they can be confounded by team style and crew turnover. New-role players should use transition priors rather than their old role average. OMEGA should prefer uncertainty to false precision.

Appendix B. Settlement and Integrity Regression Tests

Regression	Required behavior
GSIS Combined parity	For sampled games, reconstructed defensive Combined equals official GSIS Combined for every player.
FanDuel contract parity	For sampled settled T+A markets, model settlement count matches book settlement after defense + ST rules are applied.
No market leakage	No line, odds, market-implied probability or closing result is imported by the independent feature builder.
Current-game leakage	No target-game tackles, snaps, outcomes, or postgame role information enter pregame rows.
Identity fail-closed	Ambiguous player/team/game joins BLOCK rather than fuzzy-merge silently.
No-play filter	Nullified/no-play rows cannot create tackle opportunities unless final official stat explicitly credits one.
Loose-ball no-advance	No tackle event generated for loose-ball recovery without advance, matching NFL rule.
Sack inclusion	Official sack tackle credit is preserved in T+A target.
Special-teams inclusion	Book contracts that include ST produce separate, auditable ST component; contracts that exclude ST do not.
Miscellaneous exclusion	Miscellaneous tackles never leak into a defense+ST contract unless explicitly specified.
Overtime switch	OT included or excluded exactly according to settlement contract.
Correction finalization	Historical label builder uses finalized official stats and records correction metadata where available.
Venue shrinkage	Unstable venue effects shrink to zero and cannot be manually overridden to manufacture edge.
Prospective immutability	Frozen probability and market capture hashes are persisted before outcomes are known.

Appendix C. Synthetic Worked Example

The numbers in this appendix are synthetic. They illustrate the mechanics only; they are not a recommendation or an estimate for a real player.

Suppose an every-down linebacker faces an offense projected for 65.5 offensive plays. OMEGA estimates that 43.0 of those plays are tackle-eligible defensive events after accounting for incompletions, touchdowns, no-plays, and other non-credit endings. The linebacker is projected for 93% defensive exposure overall, but his event-weighted exposure is 95% because the expected play mix favors packages in which he stays on the field.

The allocation model estimates that, when he is on the field for a tackle-eligible event, he receives a combined credit 17.2% of the time after adjusting for play-type funnel, teammate competition and role. That produces approximately 7.0 defensive combined credits in expectation. His expected special-teams contribution is 0.22 because he retains a limited coverage role under the applicable book's defense+ST settlement contract. The approved grading layer contributes essentially zero after shrinkage.

$$\text{Synthetic } xT+A = 43.0 \times 0.95 \times 0.172 + 0.22 \approx 7.25$$

The Monte Carlo engine, however, does not price the wager from 7.25 alone. It draws game volume, play mix, role, credit allocation and overtime states. The resulting distribution might place 55% of mass above 6.5 but only 39% above 7.5. That is why the exact line matters as much as the player projection.

Synthetic output	Value
xT+A mean	7.25
Median	7
Expected defensive snaps	60.9
Expected tackle-eligible events	43.0
Event-weighted exposure	95%
Credit share when exposed	17.2%
Expected ST T+A	0.22
P(Over 6.5)	55% (synthetic)
P(Over 7.5)	39% (synthetic)
Trust	Would still depend on settlement, role, price and freshness gates

What makes the example “Omega”

A rolling average sees only prior tackles. OMEGA explains the count as a product of opportunity, event eligibility, exposure, matchup allocation, teammate competition, settlement-specific special teams, and uncertainty. Every term can be audited. If the market disagrees, the analyst can ask whether OMEGA is projecting more plays, a different play mix, a role change, a position funnel, or a grading effect - rather than simply staring at an unexplained black-box edge.

Appendix D. Sources and Research Notes

Sources were reviewed on 2026-09-11. Rules, house rules and data schemas can change; the production system should archive the exact version used and re-audit periodically.

[1] NFL GSIS - Guide for Statisticians, “NFL Defensive Scoring Guidelines.”

https://www.nflgsis.com/gsis/documentation/stadiumguides/guide_for_statisticians.pdf

[2] NFL GSIS - Cumulative Game Stats / StatXML report documentation.

<https://www.nflgsis.com/gsis/Documentation/Partners/StatXMLRptDocumentation.pdf>

[3] NFL Fantasy - Official Stat Corrections (NFL League Office / Elias Sports Bureau).

<https://fantasy.nfl.com/research/statcorrections>

[4] nflfastR - NFL Stats Variables; defensive tackle fields.

https://nflfastR.com/reference/nfl_stats_variables.html

[5] nflfastR - play-by-play field descriptions / fast_scraper documentation.

https://nflfastR.com/reference/fast_scraper.html

[6] nflreadr / nflverse - Snap Counts data dictionary.

https://nflreadr.nflverse.com/articles/dictionary_snap_counts.html

[7] nflreadr / nflverse - Participation data dictionary.

https://nflreadr.nflverse.com/articles/dictionary_participation.html

[8] NFL Next Gen Stats - Tackle Probability overview (2024 advanced metrics).

<https://www.nfl.com/news/next-gen-stats-new-advanced-metrics-you-need-to-know-for-the-2024-nfl-season>

[9] NFL Big Data Bowl 2024 - tackling data description.

<https://www.kaggle.com/competitions/nfl-big-data-bowl-2024/data>

[10] “Fractional tackles: leveraging player tracking data for within-play tackling evaluation in American football.”

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11739690/>

[11] FanDuel Sportsbook House Rules - American Football (Indiana page used as an example; jurisdiction-specific).

<https://www.fanduel.com/fanduel-sportsbook-house-rules-in>

[12] DraftKings Help - listed NFL Pick6 stat categories, including Solo Tackles, Assisted Tackles, and Tackles + Assists.

<https://help.draftkings.com/hc/en-us/articles/20847573983123-What-Stats-are-offered-for-Pick6-US>

[13] PFF - historical study of NFL stats-crew tackle bias. Used only to motivate a modern re-test, not as a production multiplier.

<https://www.pff.com/news/nfl-stats-crew-tackle-bias>

[14] nflverse data update / availability schedule.

https://nflreadr.nflverse.com/articles/nflverse_data_schedule.html

[15] NFL GSIS Documentation index and data-feed descriptions.

<https://www.nflgsis.com/gsis/Documentation/Default.htm>

Research interpretation notes

- NFL / GSIS sources define official statistical semantics and are treated as authoritative for tackle-credit rules.
- nflverse / nflfastR documentation defines the public data fields available to reproduce historical labels and features; the engine must still reconcile these fields against official GSIS totals.
- FanDuel rules are cited to demonstrate that at least one published T+A settlement definition includes both defense and special teams and that overtime can count. Production must verify the exact jurisdiction and current rules.
- Historical stat-crew-bias research is evidence that venue effects are worth testing, not proof that any old venue multiplier persists today.
- NFL Next Gen Stats and Big Data Bowl work support the concept that tackling should be evaluated relative to opportunity and geometry. OMEGA's public-data implementation is a separate model and should not claim to reproduce proprietary NGS probabilities.
- All “Omega Lab” ideas remain hypotheses until chronological and prospective testing shows incremental predictive value.

Closing principle

The edge, if it exists, is likely to live in the seams: between snaps and tackle-eligible events, between physical involvement and official credit, between defensive role and special-teams role, between a player and his teammates, and between the box score and the sportsbook’s exact settlement contract.

OMEGA should be built to find those seams without cheating. The model probability must exist before the market is consulted. Creative hypotheses must survive ablation. Unresolved identity, role, settlement or timestamp questions must block promotion. If the engine eventually discovers that only a handful of highly specific situations each week carry robust edge, that is a feature rather than a failure: the objective is not volume, it is correctly priced selectivity.